

ENGR 2405 – Electrical Circuits I

Laboratory #11: Simulated 3-Phase AC Circuit

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Description

This lab builds a three op-amp network that turns a single sinusoidal source into a balanced three-phase set. The generator output is phase 1 (P_1). A unity inverter (U_1) flips it by 180° , an RC section adds another 60° of lag, and a non-inverting amplifier of gain 2 (U_3) restores the amplitude the RC divider took away, producing phase 3 (P_3) at 240° of lag. The third op-amp (U_2) is an inverting summer that adds P_1 and P_3 with equal weight and inverts the result, which lands at 120° of lag and becomes phase 2 (P_2). The RC section is the only frequency-dependent part of the circuit, so a given R and C pair produces a correct three-phase set at exactly one frequency. We build nine versions of the circuit using three resistors and three capacitors, calculate the design frequency for each pair, and then measure the frequency at which the circuit actually balances along with the phase of P_2 and P_3 relative to P_1 .

Throughout the report a phase angle is quoted as a lag with respect to P_1 , so P_2 at 120° lags P_1 by 120° and P_3 at 240° lags it by 240° , which is the same signal as a 120° lead. That is the convention the handout uses.

Circuit and Design Equations

The signal chain from P_1 to P_3 is shown in Figure 1. U_1 is an inverting amplifier with $R_1 = R_2 = 10\text{ k}\Omega$, so its gain is -1 and its output is P_1 shifted by 180° . That output drives the series combination of R_{freq} and C_{freq} , and we tap the signal across the capacitor:

$$\frac{V_C}{V_{U1}} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}, \quad \left| \frac{V_C}{V_{U1}} \right| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \quad \angle = -\tan^{-1}(\omega RC).$$

Setting the lag to 60° gives the design condition

$$\tan^{-1}\left(\frac{R}{|Z_C|}\right) = \tan^{-1}(\omega RC) = 60^\circ \implies \omega RC = \tan 60^\circ = \sqrt{3} \implies \boxed{f = \frac{\sqrt{3}}{2\pi RC}}$$

At that frequency the divider magnitude is $1/\sqrt{1+3} = 1/2$, which is why U_3 is set for a gain of 2 with $R_9 = R_{10} = 10\text{ k}\Omega$. The output of U_3 is therefore the same amplitude as P_1 and lags it by $180^\circ + 60^\circ = 240^\circ$.

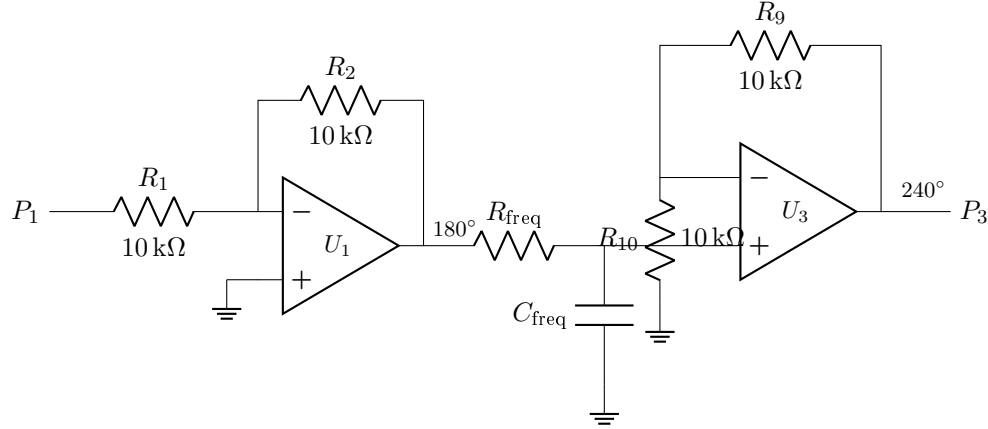


Figure 1: Phase-shift chain. U_1 inverts, the RC section adds 60° of lag and halves the amplitude, and U_3 restores the amplitude with a gain of 2.

The summing stage is shown in Figure 2. With $R_5 = R_3 = R_7 = 10\text{ k}\Omega$ the output is

$$P_2 = -\left(\frac{R_7}{R_5}P_1 + \frac{R_7}{R_3}P_3\right) = -(P_1 + P_3).$$

Writing the two inputs as phasors with $P_1 = V\angle 0^\circ$ and $P_3 = V\angle 120^\circ$ (a 240° lag is a 120° lead),

$$P_1 + P_3 = V\left(1 + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right) = V\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = V\angle 60^\circ,$$

so the sum already has the right magnitude, and the inversion built into the summer moves it to $V\angle 240^\circ = V\angle -120^\circ$. That is a 120° lag, which is P_2 . All three outputs then have the same amplitude and are spaced 120° apart, as sketched in Figure 3.

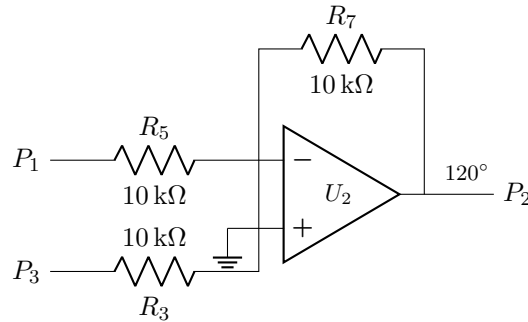


Figure 2: Summing stage. U_2 adds P_1 and P_3 with unity weight and inverts, which places P_2 at 120° of lag.

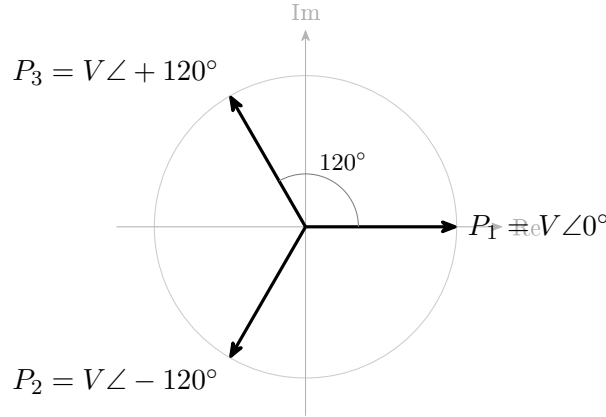


Figure 3: Phasor diagram of the three outputs at the design frequency. Reading the sequence as $P_1 \rightarrow P_2 \rightarrow P_3$, each phase lags the previous one by 120° .

Procedure

1. We picked three resistors, $1\text{ k}\Omega$, $4.7\text{ k}\Omega$ and $10\text{ k}\Omega$, and three capacitors, 100 nF , 220 nF and 470 nF , to use as R_{freq} and C_{freq} . Printed values were used for the calculations rather than LCR readings.
2. We calculated the design frequency $f = \sqrt{3}/(2\pi RC)$ for all nine R and C combinations in SciLab before building anything, so we knew where to set the generator for each trial.
3. We built the circuit on the XK700 trainer with $R_1 = R_2 = R_3 = R_5 = R_7 = R_9 = R_{10} = 10\text{ k}\Omega$. The supply rails were set to $\pm 12\text{ V}$ from the trainer, and every op-amp package got a rail connection before any signal was applied.
4. We set the function generator for a 5 V amplitude sine wave at the calculated frequency for the first RC pair and applied it at P_1 .
5. With channel 1 on P_1 and channel 2 on P_3 , we adjusted the generator frequency until the two amplitudes matched, then recorded that frequency as f_{measured} .
6. We measured the phase of P_3 with the oscilloscope time cursors. The TDS 1002 has no automatic phase readout, so we placed cursors on corresponding rising zero crossings of P_1 and P_3 , read Δt and the period T , and computed $\phi = 360^\circ \Delta t/T$.
7. We moved channel 2 to P_2 and repeated the amplitude and phase measurement at the same frequency.
8. We swapped in the next R and C pair and repeated steps 4 through 7 until all nine combinations were done.

Calculations

The design frequency for each RC pair was computed with the SciLab script below. The resistor and capacitor arrays hold the values we measured on the LCR meter, so the printed frequencies are the ones we actually set the generator to.

```

1 // ENGR 2405 Lab 11 - design frequency of the RC phase shifter
2 // The RC tap must lag by 60 deg: tan(60) = w*R*C = sqrt(3)
3 // => f = sqrt(3) / (2*pi*R*C)
4 clear;
5
6 R = [1.0e3  4.7e3  10.0e3]; // resistor values, ohms
7 C = [100e-9 220e-9 470e-9]; // capacitor values, farads
8
9 mprintf("    R (ohm)      C (F)      f_calc (Hz)\n");
10 for i = 1:length(R)
11     for j = 1:length(C)
12         f = sqrt(3) / (2*pi*R(i)*C(j));
13         mprintf("%10.1f  %10.2e  %11.2f\n", R(i), C(j), f);
14     end
15 end

```

Listing 1: SciLab program lab11_freq.sce.

R (ohm)	C (F)	f_calc (Hz)
1000.0	1.00e-07	2756.64
1000.0	2.20e-07	1253.02
1000.0	4.70e-07	586.52
4700.0	1.00e-07	586.52
4700.0	2.20e-07	266.60
4700.0	4.70e-07	124.79
10000.0	1.00e-07	275.66
10000.0	2.20e-07	125.30
10000.0	4.70e-07	58.65

Listing 2: SciLab runtime output.

The prelab table was worked out for 100 nF, 470 nF and 1 μ F. The 1 μ F parts were not available at the bench, so we substituted 220 nF and recomputed the column above before building anything.

As a check on one row by hand, with $R = 1 \text{ k}\Omega$ and $C = 100 \text{ nF}$,

$$f = \frac{\sqrt{3}}{2\pi(1000)(100 \times 10^{-9})} = \frac{1.7321}{6.2832 \times 10^{-4}} = 2756.6 \text{ Hz},$$

and back-substituting, $\omega RC = 2\pi(2756.6)(10^{-4}) = 1.7321 = \sqrt{3}$, so $\tan^{-1}(\sqrt{3}) = 60^\circ$ as required.

Each phase angle in the results table was converted from a cursor time difference using

$$\phi = 360^\circ \frac{\Delta t}{T} = 360^\circ f_{\text{meas}} \Delta t.$$

For the same row, $f_{\text{meas}} = 2700 \text{ Hz}$ gives $T = 370.4 \mu\text{s}$, and the P_3 cursor reading $\Delta t = 244 \mu\text{s}$ becomes $\phi = 360(244)/370.4 = 237.2^\circ$.

Data and Results

Table 1 lists all nine trials. The handout's example table asks for $\phi(P_1)$, but P_1 is the reference for every phase measurement and sits at 0° by definition, so we recorded $\phi(P_3)$ in that column instead.¹ Phases are lags with respect to P_1 ; the design targets are 240° for P_3 and 120° for P_2 . The raw cursor readings are included so the angles can be traced back to what was on the screen.

¹ $\phi(P_3)$ is what the procedure actually asks to measure, so we read the column heading as a typo.

Table 1: Measured results for all nine RC combinations. Phase angles are computed from the cursor readings as $\phi = 360^\circ f_{\text{meas}} \Delta t$.

R (k Ω)	C (nF)	f_{calc} (Hz)	f_{meas} (Hz)	% err	Δt_{P_3} (ms)	$\phi(P_3)$ ($^\circ$)	Δt_{P_2} (ms)	$\phi(P_2)$ ($^\circ$)
1.0	100	2756.64	2700	-2.05	0.244	237.2	0.124	120.5
1.0	220	1253.02	1200	-4.23	0.540	233.3	0.260	112.3
1.0	470	586.52	586	-0.09	1.120	236.3	0.580	122.1
4.7	100	586.52	586	-0.09	1.140	240.5	0.560	118.1
4.7	220	266.60	266	-0.23	2.520	241.3	1.240	118.8
4.7	470	124.79	125	+0.17	5.300	238.5	2.600	117.0
10.0	100	275.66	276	+0.12	2.400	238.5	1.200	119.2
10.0	220	125.30	125	-0.24	5.400	243.0	2.600	117.0
10.0	470	58.65	59	+0.59	11.400	242.1	5.700	121.1

Table 2 pulls out how far each angle sits from its target. It also lists the angle an ideal RC section would produce at the frequency we actually ran at, which separates the part of the phase error that follows from being off the design frequency from the part that does not.

Table 2: Phase deviation from the design targets of 240° and 120° . The ideal column is $180^\circ + \tan^{-1}(\omega RC)$ evaluated at f_{meas} , which is what P_3 should read given that we were not exactly on the design frequency.

R (k Ω)	C (nF)	$\phi(P_3)$ ideal ($^\circ$)	$\phi(P_3)$ meas. ($^\circ$)	dev. ($^\circ$)	$\phi(P_2)$ meas. ($^\circ$)	dev. ($^\circ$)
1.0	100	239.48	237.2	-2.8	120.5	+0.5
1.0	220	238.92	233.3	-6.7	112.3	-7.7
1.0	470	239.98	236.3	-3.7	122.1	+2.1
4.7	100	239.98	240.5	+0.5	118.1	-1.9
4.7	220	239.94	241.3	+1.3	118.8	-1.2
4.7	470	240.04	238.5	-1.5	117.0	-3.0
10.0	100	240.03	238.5	-1.5	119.2	-0.8
10.0	220	239.94	243.0	+3.0	117.0	-3.0
10.0	470	240.15	242.1	+2.1	121.1	+1.1
Mean deviation				-1.0		-1.5
RMS deviation				3.1		3.2

Observations & Discussion

The measured and calculated frequencies agreed closely for seven of the nine combinations, all within 0.6% of the design value and six of them within 0.25%. The two outliers are the two highest-frequency trials, 1 k Ω with 100 nF (-2.05%) and 1 k Ω with 220 nF (-4.23%), and in both the recorded frequency is a round number, 2700 Hz and 1200 Hz. Those are readout results rather than physical ones. Above about 1 kHz we were reading the generator to the nearest 100 Hz, so the discrepancies of 57 Hz and 53 Hz are roughly half a step in each case. The remaining seven rows sat

where the display resolution was fine enough to land on the calculated value, and they did.

That column is a weaker test than it looks, and it is worth being honest about that. We set f_{meas} by adjusting the generator until the amplitude of P_3 matched P_1 , and with the gain of 2 from U_3 included the balance condition is

$$A(f) = \frac{2}{\sqrt{1 + (\omega RC)^2}}, \quad \omega RC = \sqrt{3} \frac{f}{f_0},$$

which gives $A = 0.964$ at 5% above f_0 and $A = 0.929$ at 10% above. On a 5 V amplitude signal that is only 180 mV and 355 mV of change, so matching the two traces by eye pins the frequency to something like $\pm 5\%$. The phase reading is no sharper on its own: differentiating $\phi_{RC} = \tan^{-1}(\sqrt{3}f/f_0)$ at $f = f_0$ gives about 0.25° per 1% of frequency error against a cursor resolution of one to two degrees. The sub-percent agreement in the frequency column is therefore not evidence that the model is accurate to a fraction of a percent. What the two criteria establish together is a consistency check: the amplitudes balance and all three phases sit near their targets at the same frequency, and the model predicts that frequency to within the resolution of both.

The phases themselves came out well. $\phi(P_3)$ landed within 3.7° of 240° in eight of nine trials and $\phi(P_2)$ within 3.0° of 120° in eight of nine, with RMS deviations of 3.1° and 3.2° , about 1% of a full cycle. Both means are slightly negative, -1.0° and -1.5° , so there is a small systematic shortfall in lag on top of the scatter. Part of that is not an error at all: if $f < f_0$ then $\omega RC < \sqrt{3}$ and the RC section lags by less than 60° , so P_3 should come in under 240° . Table 2 evaluates $180^\circ + \tan^{-1}(\omega RC)$ at the frequency we actually used, and for the 1 k Ω and 220 nF row that ideal value is 238.9° , which accounts for about 1.1° of the 6.7° gap. That same row has by far the worst $\phi(P_2)$ at 112.3° , which follows, since P_2 is assembled from P_1 and P_3 by the summer and inherits whatever error P_3 carries. It is the only row where we were meaningfully off the design frequency and the only row where both angles are more than 5° out.

The rest of the scatter comes from resolution and from the parts. The TDS 1002 has no automatic phase function, so every angle came from placing two cursors on corresponding zero crossings and reading Δt against the period. Our Δt readings carry two or three significant figures, and one unit in the last figure is already worth about a degree: 1 μs out of a 370 μs period is 0.97° in the first row, and 0.1 ms out of 16.9 ms is 2.1° in the last. That alone brackets most of what we saw. On the component side, f_0 depends on the product RC , so a tolerance in either part propagates one-for-one into the calculated frequency, and 5% resistors with 10% capacitors put a worst case band of roughly 15% around each nominal value. We used printed values rather than LCR readings, so entries in the f_{calc} column could be off by more than the differences under discussion. The analysis also treats the capacitor node as unloaded when it actually drives the non-inverting input of U_3 , whose input capacitance and the breadboard together add a few picofarads. That matters most for the 100 nF trials at the top of our range and is negligible at 470 nF. Each op-amp stage contributes a small lag of its own at finite loop gain as well, and those accumulate through the chain, consistent with the mean deviation in $\phi(P_2)$ being slightly larger in magnitude than the one in $\phi(P_3)$.

As for where a circuit like this gets used, the point of it is producing a balanced three-phase reference from a single-phase source and a handful of parts. On a bench that lets us exercise three-phase equipment such as phase sequence indicators, protective relays, or the front end of a three-phase rectifier without needing an actual three-phase supply. In a control setting the same three signals can serve as the reference for the gate drive of a three-phase inverter driving an induction motor, where the reference frequency sets the motor speed, and the same idea appears in signal processing where fixed phase offsets are generated this way for quadrature modulation. The limitation is the one these nine trials demonstrate directly: the shift is correct at a single frequency for a given R and C , so a variable-frequency version needs R or C to track the frequency instead of

staying fixed. That is why the practical modern approach is a digital one, where three phases are read out of a lookup table clocked at whatever rate the application needs and the 120° spacing is exact by construction.